

Richard A. Matos Arderí

COMPUTER SCIENTIST · DATA SCIENCE, ML & AI · FULL-STACK DEVELOPER

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LANGUAGES

Spanish Native

English C1 · Advanced

French A2 · Elementary

HONORS & AWARDS

Scientific Merit Prize

Cuban Higher Education · 2026

1st Place · Honors Examination in Ordinary Differential Equations

MATCOM, UH · 2026

“Relevant” paper at BIOQUIM 2025

Hybrid modeling of leptospirosis with vaccination

RESEARCH INTERESTS

Biomedical AI

Epidemiological Modeling

Vaccine Development

Mathematical Biology

Distributed Systems

PROFILE

Computer Scientist trained at MATCOM (University of Havana), specializing in Data Science, Machine Learning and Artificial Intelligence, with strong full-stack and web development experience. Awarded the Scientific Merit Prize for an outstanding research record. Currently a Consultant Specialist at the Finlay Vaccine Institute (BioCubaFarma), where I design AI-powered platforms — multi-agent systems, RAG pipelines and AutoML workflows — and ship web, mobile and desktop applications. I take products end to end, from data and mathematical modeling to polished, deployable software across web, Android, iOS and desktop.

4.71

Academic GPA / 5.00

6+

Publications & talks

3

Awards & distinctions

24

Software projects

TECHNICAL SKILLS

Programming Languages

Python · Rust · TypeScript / JavaScript · C · C++ · C# · R · Haskell · Prolog · SQL · LaTeX

Data Science & ML

scikit-learn · PyTorch · TensorFlow · XGBoost · LightGBM · CatBoost · SHAP · LIME · MLflow · NumPy · Pandas · SciPy · SymPy

AI / Agents / NLP

LangChain · LangGraph · RAG · Multi-agent systems · AutoML (FLAML) · n8n · HuggingFace · Ollama · ChromaDB · Ontologies & SPARQL (rdflib)

Modeling & Simulation

ODE / SEIR models · Agent-based modeling (Mesa) · Cellular automata · Markov chains (CTMC) · Metaheuristics (GA, SA, Tabu, GRASP) · Sensitivity analysis (SALib/Sobol)

Databases

PostgreSQL · MySQL · SQL Server · MongoDB · Redis · SQLAlchemy / Alembic

Web & Full-Stack

React · Next.js · Node.js · FastAPI · Django · Tailwind CSS · REST APIs · WebSockets

Mobile & Desktop

Flutter · Android · iOS · React Native · Electron / Tauri

Big Data & Distributed

Hadoop · HDFS · PySpark · Celery · Redis · Distributed crawling & indexing

DevOps & Tools

Docker · Docker Compose · Git · GitHub Actions (CI/CD) · CodeQL · Linux

QA & Testing

pytest · unittest · Selenium · Cypress · Playwright · Postman · JMeter · TDD · SonarQube · ruff / black / mypy

PROFESSIONAL EXPERIENCE

Consultant Specialist Mar 2025 – Present
Finlay Vaccine Institute (IFV) · BioCubaFarma
Data-science and AI consulting for vaccine development: epidemiological modeling, clinical-data integration and auditing, and the design of predictive models and supporting software within IFV and BioCubaFarma projects.

FEATURED PROJECTS

- FinIAI** AI · AutoML · Health
AI platform for optimizing vaccine development through integrated data management at the Finlay Vaccine Institute: multi-agent systems, RAG pipelines and AutoML workflows over clinical-trial data.
- AuditSoft** Audit · Clinical data
Software for auditing clinical-trial data, ensuring quality, traceability and integrity of information within the IFV regulatory framework. Presented at COMPUMAT 2025.
- Simmunity** ABM · ODE · FastAPI · Distributed
Computational simulator of the human immune system: agent-based, ODE and stochastic (Gillespie) models, a FastAPI REST layer, and distributed computing with Celery/Redis.
- climaXtreme** PySpark · HDFS · Docker
Analysis and modeling of extreme climate events over 8.6M+ records with Hadoop and PySpark; Streamlit dashboard, ML predictions and full Docker deployment.
- DistriSearch** Python · Distributed
Distributed search engine with distributed crawling, indexing and query processing (Distributed Systems final project).

CONFERENCE CONTRIBUTIONS

- “Hybrid Modeling: A system for epidemiological modeling”**
XLI Scientific Conference, Faculty of Physics, UH · May 2026
- “AuditSoft: Software for auditing clinical-trial data”**
COMPUMAT 2025 · International
- “ODE Mathematical Model for leptospirosis transmission”**
COMPUMAT 2025 · International
- “Hybrid Modeling of leptospirosis transmission with vaccination strategies”**
BIOQUIM 2025 · National · “Relevant” distinction
- “Interactive Web Application for predicting Acute Myocardial Infarction”**
BIOQUIM 2025 · National · with A. Ponce González
- “ML models for in-hospital AMI mortality prediction and inverse optimization of clinical variables”**
BIOQUIM 2025 · National · with A. Ponce González, A. Romero Imbert

EDUCATION

- B.Sc. in Computer Science** Jan 2023 – Jun 2026
Faculty of Mathematics and Computer Science (MATCOM), University of Havana
GPA: 4.71 / 5.00 · Cohort C411.
- Diploma in Clinical Trials** 2025
Center for Genetic Engineering and Biotechnology (CIGB) · BioCubaFarma
Specialized postgraduate training in the design, conduct, regulation and data management of clinical trials.
- International Winter School** 2026
University of Havana
Intensive program of international scientific training and exchange.

- Hybrid Modeling** ODE · ML · Epidemiology
System for epidemiological modeling combining differential-equation models with machine learning; applied to leptospirosis transmission and the evaluation of vaccination strategies.
- ML Studio** AutoML · Predictive models
Machine-learning modeling environment for predicting in-hospital mortality from AMI and performing inverse optimization of clinical variables, with interpretability (SHAP) and validation.
- Clinical Trial DB Design** PostgreSQL · ER Modeling
Relational database design (13+ tables, 3NF) for the ISCANI clinical trial of the VCN7-T vaccine: patients, randomization, adverse events and regulatory audit trail.
- HULK-Compiler-RS** Rust · Compilers
A complete compiler for the HULK language in Rust: lexical, syntactic and semantic analysis, and garbage collection, focused on performance and safety.

RESEARCH PROJECTS

- Optimizing vaccine development through integrated data management and AI**
Finlay Vaccine Institute (IFV) · BioCubaFarma
- Data Ecosystem Evolution Project**
DICEI · IFV · BioCubaFarma
- Development of a novel vaccine against Leptospirosis**
IFV · BioCubaFarma
- AMI registry with variable homogenization and quality indicators in Cuba**
“Dr. Ernesto Guevara de la Serna” Teaching Hospital · MINSAP
- VLIR-UOS: Integrated Real-Time Dengue Surveillance in Cuba for Outbreak Prediction**
VLIR-UOS international cooperation